

kfjtsw7og

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[12]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from scipy.cluster.hierarchy import linkage, dendrogram
from sklearn.cluster import AgglomerativeClustering
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[14]: # Step 1: Load data
data = pd.read_csv("C:/Users/cool/Desktop/ML ass/cars_clus.csv")
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[18]: # Step 2: Preprocess

# Drop non-numeric columns
data_clean = data.drop(['manufact', 'model', 'type', 'partition'], axis=1)

# Replace '$null$' with np.nan
data_clean.replace('$null$', np.nan, inplace=True)

# Convert all columns to numeric (some may still be object because of '$null$')
data_clean = data_clean.apply(pd.to_numeric)

# Drop rows with missing values
data_clean = data_clean.dropna()

# Scale the data
scaler = StandardScaler()
data_scaled = scaler.fit_transform(data_clean)
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[20]: # Step 3: Agglomerative Clustering and Dendrograms

linkage_methods = ['single', 'complete', 'average']

plt.figure(figsize=(20, 6))

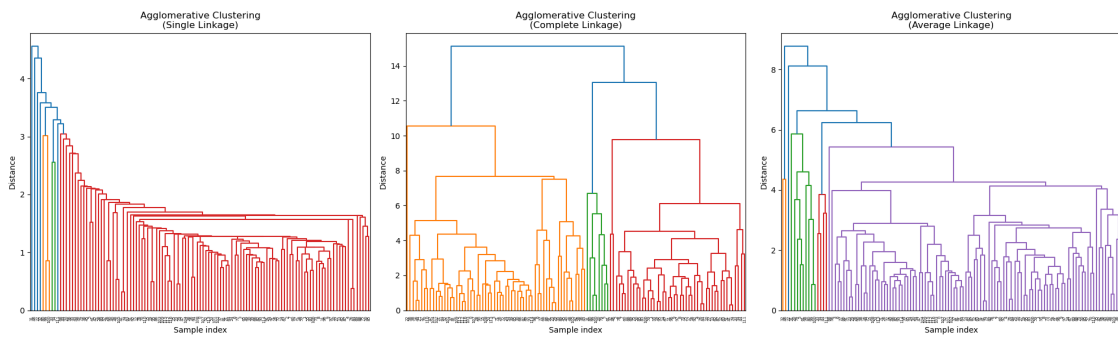
for i, method in enumerate(linkage_methods):
    plt.subplot(1, 3, i + 1)
    linked = linkage(data_scaled, method=method)
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dendrogram(linked,
            orientation='top',
            distance_sort='ascending',
            show_leaf_counts=True)
plt.title(f'Agglomerative Clustering\n({method.capitalize()} Linkage)')
plt.xlabel('Sample index')
plt.ylabel('Distance')

plt.tight_layout()
plt.show()

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